

ASCIA OS

Hardware Ecosystem & Device Compatibility Guide

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This document provides the complete hardware ecosystem supported by ASCIA OS, including third-party partner brands currently integrated, and ASCIA's own proprietary hardware line currently in development. It is intended for development agencies to understand the full scope of device compatibility, protocol coverage, and integration requirements.

All third-party devices listed below are available for physical testing at the ASCIA development lab in Montreal. ASCIA proprietary devices are currently in production and will follow the protocols described — agencies are encouraged to provide firmware guidance and integration recommendations to ensure full ecosystem compatibility and stability.

ASCIA proprietary devices take priority in discovery reliability, stability, and feature completeness.

SECTION 1 — Third-Party Partner Brands

ASCIA OS currently integrates with the following partner brands and ecosystems. All devices from these brands are available for testing at the Montreal development lab unless otherwise noted.

Brand	Category	Primary Protocol(s)
Aqara	Sensors, Presence, Cameras	Zigbee, Wi-Fi
Ring	Doorbells, Cameras, Alarm	Wi-Fi
Philips Hue	Lighting, Controls	Zigbee, Wi-Fi
Konnected	Alarm Panel, Wired Sensors	Wi-Fi, Ethernet
Zemismart	Curtains, Switches, Relays	Zigbee, Wi-Fi
Sonoff	Switches, Relays, Sensors	Zigbee, Wi-Fi
Linpotech	Presence Sensors	Wi-Fi
Reolink	Cameras	ONVIF, RTSP, Wi-Fi, PoE
Roborock	Robot Vacuums	Wi-Fi
Everything Smart	Presence Sensor One	Wi-Fi, PoE
Govee	Lighting, LED Strips	Wi-Fi, BLE
Nest	Thermostats, Cameras	Wi-Fi
Ecobee	Thermostats	Wi-Fi
Sinope	Thermostats, Plugs	Zigbee, Wi-Fi
Moes	Switches, Thermostats, Relays	Zigbee, Wi-Fi
Shelly	Relays, Switches, Sensors	Wi-Fi, MQTT
TP-Link / Omada	Network Infrastructure	Ethernet, PoE, Wi-Fi

Brand	Category	Primary Protocol(s)
Ubiquiti / UniFi	Network Infrastructure	Ethernet, PoE, Wi-Fi
Worx	Robot Lawnmowers	Wi-Fi
iRobot	Robot Vacuums	Wi-Fi
Dreametech	Robot Vacuums	Wi-Fi
Litter-Robot	Smart Litter Boxes	Wi-Fi
Moen	Smart Shutoff, Faucets	Wi-Fi, Zigbee
Rachio	Smart Sprinkler Controllers	Wi-Fi
Ondilo ICO	Pool / Spa Sensors	Wi-Fi
Honeywell	Thermostats, Sensors	Wi-Fi, Zigbee
Sonos	Speakers, Audio	Wi-Fi, Ethernet
Withings	Health Sensors	Wi-Fi
ABB KNX	Full KNX Catalog	KNX (TP)
MDT KNX	Full KNX Catalog	KNX (TP)
SMLIGHT	Zigbee Coordinators	Zigbee, Ethernet
Eero	Wi-Fi Mesh Network	Wi-Fi, Ethernet

SECTION 2 — Sensors

Sensor Type	Connectivity	Protocol(s)
Motion Sensor	Wireless / Wired	Zigbee, Wired to Konnected Alarm Panel
Contact Sensor (Door/Window)	Wireless / Wired	Zigbee, Wired to Konnected Alarm Panel
Temperature & Humidity Sensor	Wireless	Zigbee
Vibration Sensor	Wireless	Zigbee
Pool / Jacuzzi State Sensor	Wireless	Wi-Fi
Water Leak Sensor	Wireless	Zigbee
Pressure Sensor	Wireless	Zigbee
mmWave Presence Sensor	Wired	Wi-Fi, PoE
Sleep Sensor	Wired	Wi-Fi
CO2 / Smoke Sensor	Wireless / Wired	Zigbee, Wired to Konnected Alarm Panel
Glass Break Sensor	Wireless / Wired	Zigbee, Wired to Konnected Alarm Panel

SECTION 3 — Lighting

Device Type	Connectivity	Protocol(s)
LED Controller	Wired	Zigbee
LED Strip	Wired / Wireless	Zigbee, Wi-Fi
Smart Bulb	Wired	Zigbee, Wi-Fi
Spotlight / Spot	Wired	Zigbee, Wi-Fi
Downlight	Wired	Zigbee, Wi-Fi
Decorative Lamp (Floor, Table, etc.)	Wired	Zigbee, Wi-Fi
Outdoor Lighting (Bollard, Wall, Exterior LED)	Wired	Zigbee, Wi-Fi
Pool Lighting	Wired	Zigbee, Wi-Fi
Dynamic TV Lighting (Philips Hue Play)	Wired	Zigbee, Wi-Fi

SECTION 4 — Wall Switches & Relays

Device Type	Connectivity	Protocol(s)
Universal Relay (Curtains, Garage, Gate, Switch, Blind, Plug, Appliances)	Wired	Wi-Fi, Zigbee
Standard Switch	Wired	Wi-Fi, Zigbee
Dimmer Switch	Wired	Wi-Fi, Zigbee
NS Panel (Display Switch)	Wired	Wi-Fi, Zigbee
Tap Dial	Wireless	Zigbee
Tap Switch	Wireless	Zigbee
Multi-Button Switch / Scene Switch	Wireless	Zigbee, Wi-Fi

SECTION 5 — Cameras & Video

Device Type	Connectivity	Protocol(s)
Outdoor Camera	Wired / Battery	ONVIF, RTSP, PoE, Wi-Fi
Indoor Camera	Wired / Battery	ONVIF, RTSP, PoE, Wi-Fi
Video Doorbell	Wired / Battery	ONVIF, RTSP, PoE, Wi-Fi

SECTION 6 — Robots & Automated Devices

Device Type	Connectivity	Protocol(s)
Robot Vacuum (with/without base)	Wireless	Wi-Fi

Device Type	Connectivity	Protocol(s)
Robot Lawnmower (with/without base)	Wireless	Wi-Fi
Pool Cleaning Robot	Wired / Wireless	Wi-Fi
Robot Litter Box	Wired	Wi-Fi
Automatic Pet Feeder	Wired	Wi-Fi, Zigbee

SECTION 7 — Curtains & Blinds

Device Type	Connectivity	Protocol(s)
Curtain Motor	Wired / Wireless	Zigbee, Wi-Fi
Blind / Shutter Motor	Wired / Wireless	Zigbee, Wi-Fi

SECTION 8 — Thermostats & Climate Control

Device Type	Connectivity	Protocol(s)
Electric Baseboard Thermostat	Wired	Zigbee, Wi-Fi
Heated Floor Thermostat	Wired	Zigbee, Wi-Fi
Centralized Thermostat (HVAC)	Wired	Zigbee, Wi-Fi
Air Conditioner / Mini-Split	Wired	Wi-Fi
Air Purifier	Wired	Wi-Fi

SECTION 9 — Smart Home Devices

Device Type	Connectivity	Protocol(s)
Smart Water Shutoff Valve	Wired	Zigbee, Wi-Fi
Smart Shower Valve	Wired	Wi-Fi
Smart Plug (Indoor)	Wired / Wireless	Wi-Fi, Zigbee
Smart Plug (Outdoor)	Wired / Wireless	Wi-Fi, Zigbee
Dome Siren / Alarm Siren	Wired / Wireless	Zigbee, Wi-Fi
Tablet Dashboard	Wired / Wireless	PoE, Wi-Fi
Smart Sprinkler Controller	Wired	Wi-Fi
Panic Button	Wired / Wireless	Wi-Fi, Zigbee
Smart Alarm Panel	Wired	Wi-Fi, PoE
Smart Garage Door Opener	Wired	Wi-Fi, Zigbee
Philips Hue HDMI Sync Box	Wired	Wi-Fi

SECTION 10 — Audio & Sound Systems

Device Type	Connectivity	Protocol(s)
Outdoor Speaker	Wired / Wireless	Wi-Fi, Ethernet
Indoor Speaker	Wired / Wireless	Wi-Fi, Ethernet
In-Ceiling / In-Wall Speaker	Wired	Wi-Fi, Ethernet
Architectural Speaker	Wired	Wi-Fi, Ethernet
Sonos Port	Wired	Wi-Fi, Ethernet
Sonos Amp	Wired	Wi-Fi, Ethernet
Subwoofer	Wired	Wi-Fi, Ethernet
Soundbar	Wired	Wi-Fi, Ethernet

SECTION 11 — Multimedia & Entertainment

Device Type	Connectivity	Protocol(s)
Amazon Echo Hub Max	Wired	Wi-Fi
Amazon Echo Dot	Wired	Wi-Fi
Amazon Echo Show	Wired	Wi-Fi
Google Home Mini	Wired	Wi-Fi
Alarm Keypad	Wireless / Wired	Zigbee, Wi-Fi
Chromecast	Wired	Wi-Fi
Apple TV	Wired	Wi-Fi, Ethernet
Fire TV Stick	Wired	Wi-Fi
LG Smart TV	Wired / Wireless	Wi-Fi, Ethernet
Samsung Smart TV	Wired / Wireless	Wi-Fi, Ethernet
Projector	Wired	Wi-Fi, Ethernet

SECTION 12 — Network Infrastructure

System	Type	Protocol(s)
TP-Link Omada (EAPs, Switches, Gateways, OC200 Controller)	Managed Network	Ethernet, PoE, Wi-Fi, SDN API
Ubiquiti UniFi (APs, Switches, Gateways, Dream Machine)	Managed Network	Ethernet, PoE, Wi-Fi, SDN API
Eero Wi-Fi Mesh	Wi-Fi Mesh	Wi-Fi, Ethernet
SMLIGHT Zigbee Coordinators (SLZB-06M and variants)	Zigbee Coordinator	Zigbee 3.0, Ethernet, Wi-Fi

SECTION 13 — KNX Systems (ABB & MDT)

ASCIA OS supports the complete KNX catalog from both **ABB KNX** and **MDT KNX**, covering all modules, actuators, switches, sensors, and accessories available in their product lines. This includes but is not limited to: lighting actuators, blind/shutter actuators, HVAC actuators, scene controllers, push-button interfaces, binary inputs, logic modules, IP routers and gateways, and energy meters.

KNX integration is handled via the KNX IP interface or KNX IP router connected to the local network. ASCIA OS communicates with the KNX bus using the standard KNX/IP protocol (KNXnet/IP Core, KNXnet/IP Tunneling, KNXnet/IP Routing).

SECTION 14 — ASCIA Proprietary Hardware (In Development)

The following devices are ASCIA's own proprietary hardware currently in development. These devices are not yet available for physical testing but follow the protocols described below. Development agencies are expected to:

1. Treat ASCIA proprietary devices as **first-priority** in discovery, stability, and feature completeness.
2. Provide firmware-level guidance and recommendations where applicable to ensure seamless integration.
3. Flag any integration requirements or constraints early so ASCIA hardware engineering can incorporate them.
4. Architect the integration layer to be hardware-agnostic where possible, so ASCIA devices can be swapped in as drop-in replacements for equivalent third-party devices on the same protocol.

ASCIA Device	Description	Protocol(s)	Status
ASCIA All-In-One Sensor	Single multi-sensor device combining: Temperature, Humidity, Air Quality, CO2, mmWave static presence, mmWave individual count & tracking, Lux, BLE interceptor (ESP32), Microphone for local AI voice activation	Wi-Fi, PoE (Ethernet)	In Development Specs Partial
ASCIA Local Server	Preconfigured mini-PC with ASCIA OS. Specs: Intel i7, 32GB DDR4, optional GPU (if AI enabled), 1TB NVMe (system), 2TB NVMe (history), 4-16TB (camera & backup), integrated network card and Zigbee coordinator (SMLIGHT-06M equivalent)	Ethernet, Wi-Fi, Zigbee 3.0	In Development Specs Partial
ASCIA Alarm Panel	Same function as Konnected Alarm Panel — supports wired sensors (22/2 cable), wired motion, door, glass break, thermal sensors, panic button integration	Ethernet, Wi-Fi	In Development
ASCIA KNX Catalog	Complete KNX product line including all actuators, switches, sensors, and accessories — full catalog equivalent to ABB/MDT	KNX (TP)	Planned
ASCIA Zigbee Device Catalog	Full catalog of proprietary Zigbee sensors and actuators across all categories	Zigbee 3.0	In Development
ASCIA PoE Switch / Scene Controller	PoE or Ethernet-based switch and scene controller for wall installation	PoE, Ethernet	In Development
ASCIA Tablet Dashboard	Custom tablet dashboard optimized for ASCIA OS — wall-mountable	PoE, Wi-Fi, Ethernet	In Development
ASCIA PoE Actuator Modules	DIN-rail actuator modules for lighting, switches, blinds, curtains — PoE powered, similar to KNX actuators but PoE-based	PoE, Ethernet	In Development
ASCIA Audio Amplifier	Multi-zone amplifier adapted for ASCIA OS — similar to Triad Audio but fully integrated with ASCIA platform	Wi-Fi, Ethernet	In Development
ASCIA Wired Alarm Sensors	Wired motion, door contact, glass break, and thermal sensors (22/2 cable) — compatible with ASCIA Alarm Panel	Wired (22/2)	In Development
ASCIA PoE LED Strip	PoE-powered LED light strip for architectural and ambient lighting	PoE, Ethernet	In Development
ASCIA Multifunction Relay	Ethernet-connected multifunction relay for universal device control	Ethernet	In Development

ASCIA Device	Description	Protocol(s)	Status
ASCIA Multifunction Thermostat	Ethernet-connected multi-zone thermostat	Ethernet	In Development
ASCIA Door Lock	Smart door lock with PoE or Ethernet connectivity	PoE, Ethernet	In Development
ASCIA Mini Dashboard (NS Panel equivalent)	Small wall-mounted display and scene controller	PoE, Ethernet, Wi-Fi	In Development
ASCIA Wall Switch (all types)	Full line of wall switches with Ethernet connectivity — all form factors and configurations	Ethernet	In Development

SECTION 15 — Notes for Development Agencies

Testing Environment	All third-party devices listed in Sections 1–13 are physically available at the ASCIA development lab in Montreal. Remote access to the lab is provided via Tailscale after NDA signing and contract start.
Mock API for Development	ASCIA will provide real Omada API JSON response samples for mock development. Agencies should build a local mock environment for daily development — the physical lab is for integration testing and validation, not for running unit tests on every commit.
ASCIA Proprietary Devices	ASCIA proprietary devices (Section 14) are in development. Agencies should architect integration layers using protocol-level abstraction so that ASCIA devices can be integrated as first-class citizens when hardware becomes available. If firmware recommendations are needed, flag them early.
Protocol Priority for MVP	The following protocols are mandatory for MVP: Zigbee (via Zigbee2MQTT), ONVIF/RTSP (cameras), MQTT, Wi-Fi (mDNS/SSDP), PoE device identification via Omada API, WireGuard VPN. KNX and Z-Wave are Phase 2.
KNX	KNX integration requires experience with ETS configuration, group address mapping, and KNXnet/IP tunneling. Both ABB and MDT catalogs must be supported. Agencies should confirm KNX experience explicitly.
Zigbee Coordinator	ASCIA uses SMLIGHT coordinators (SLZB-06M and equivalent) connected over Ethernet. Zigbee2MQTT runs as a Docker add-on. The coordinator is not USB — it is network-connected. Agencies must account for this in their add-on architecture.
Discovery Priority	ASCIA proprietary devices must be detected and onboarded with the highest priority and reliability in the discovery engine. All other brands follow standard protocol-based discovery.